

UECM3993 PRACTICAL QUIZ MARKING GUIDE

Name: _____ Student ID: _____ Marks: _____ /12

COURSE CODE & COURSE TITLE: UECM3993/EECM3993 PREDICTIVE MODELLING
 FACULTY: LKC FES, UTAR COURSE: AM, AS, FM
 SESSION: FEB 2026 LECTURER: LIEW HOW HUI

Instruction: Answer all questions in the space provided. **If you do not write your answer in the space provided, you will get ZERO mark.** An answer without necessary working steps may also receive ZERO mark.

CO4: Demonstrate supervised and unsupervised learning with statistical software

1. Write down the return value of the R command by rounding them to 5 decimal places:
`(x-min(x))/(max(x)-min(x))` when `x = c(88,53,58,18,16,87)**2`. (1.5 marks)

Ans. `x = 7744, 2809, 3364, 324, 256, 7569`

`1, 0.34095, 0.41506, 0.00908, 0, 0.97663` [1.5 marks]

2. Write down the return value of the R command:

`c(4,3) * (c(3,6,2) + c(43,21,31,46,35,41,29,25,33,40,50,36))`. (0.5 mark)

Ans. `184 81 132 147 164 129 128 93 140 129 224 114`

..... [0.5 mark]

3. Write down the return value of the R command: `table(paste("A", c(20, 20, 28, 28, 20, 20, 28, 20, 28, 20, 20, 20), sep=''), paste("C", c(13, 22, 13, 13, 12, 17, 13, 12, 13, 12, 17, 13, 17), sep=''))` .

Note that R sort labels in dictionary order. (1 mark)

Ans.

```

      C12 C13 C17 C22
A20    2   2   3   1
A28    1   4   0   0
    
```

..... [1 mark]

4. If `"X = matrix(c(29,22,41,29,36,33,40,33,18,49,35,23), 3,4)"`, write down the values of X (in matrix form) as well as the return value of the R command `apply(X, 2, sum)` (1 mark)

Ans.

```

      [,1] [,2] [,3] [,4]
[1,]   29   29   40   49
[2,]   22   36   33   35
[3,]   41   33   18   23
    
```

`[1] 92 98 91 107` [0.5 mark]

..... [0.5 mark]

5. You have downloaded life expectancy data from the United Nation World Population Prospects (2024). By using the R statistical software and the commands below:

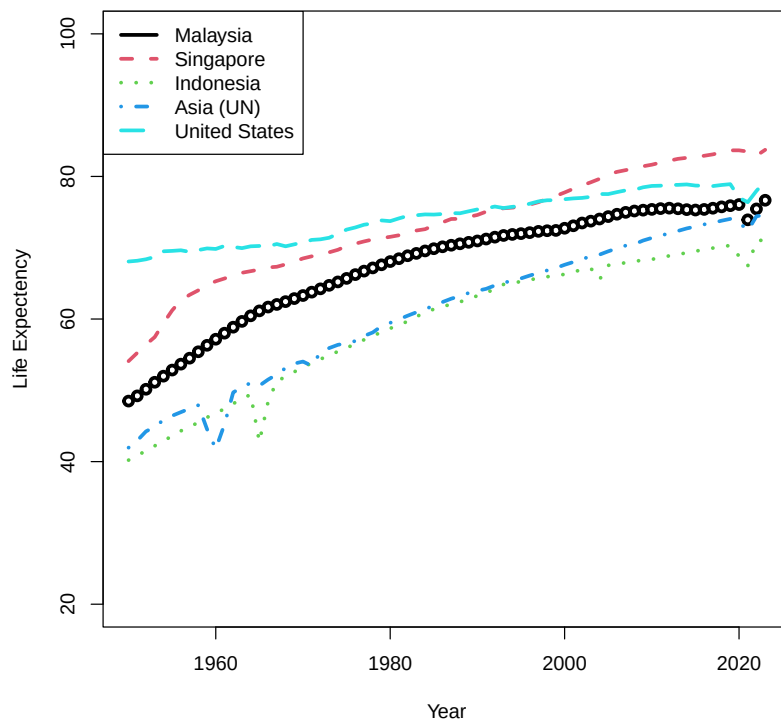
```
zipdata = "life-expectancy-unwpp.zip"
d.f = read.csv(unz(zipdata, "life-expectancy-unwpp.csv"))
print(summary(d.f))
```

You have generated a statistical summary below.

Entity	Code	Year	Life.Expectancy..age.0
Length:18722	Length:18722	Min. :1950	Min. :10.99
Class :character	Class :character	1st Qu.:1968	1st Qu.:55.99
Mode :character	Mode :character	Median :1986	Median :66.43
		Mean :1986	Mean :63.78
		3rd Qu.:2005	3rd Qu.:72.83
		Max. :2023	Max. :86.37

Demonstrate your unsupervised learning skills that you learn from Practical 1 to Practical 3 by writing down the R commands for

- selecting the five entities "Malaysia", "Singapore", "Indonesia", "Asia (UN)", "United States" from first, third and fourth columns of `d.f` and store them to `d.f.selected`; (1 mark)
- plotting the life expectancy against year for each entity in `d.f.selected` giving the following diagram. (1 mark)



Ans. A sample R script is listed below.

```
# part (i) : [1 mark]
idx1=d.f$Entity=="Malaysia"
```

```

idx2=d.f$Entity=="Singapore"
idx3=d.f$Entity=="Indonesia"
idx4=d.f$Entity=="Asia (UN)"
idx5=d.f$Entity=="United States"
d.f.selected = d.f[ idx1|idx2|idx3|idx4|idx5 , c(1,3,4)]

# part (ii) : [1 mark]
entities = c("Malaysia", "Singapore", "Indonesia",
             "Asia (UN)", "United States")
plot(d.f.selected[d.f.selected$Entity==entities[1],2:3],
     xlab="Year", ylab="Life Expectancy", lwd=3, ylim=c(20,100))
for (i in 2:length(entities)) {
  lines(d.f.selected[d.f.selected$Entity==entities[i],2:3], lty=i,
        col=i, lwd=3)
}
legend("topleft", entities, lty=1:5, lwd=3, col=1:5)

```

6. Given the training data with features **Income**, **Limit**, **Cards**, and the response **Balance** in Table 3.1.

Income	Limit	Cards	Balance
71.408	7114	2	872
98.515	8760	5	1230
16.479	5435	2	937
104.483	7140	2	583
27.578	2531	1	0
53.217	4943	2	382
15.866	3085	1	136
11.808	1300	3	0
35.61	6135	4	992
15.602	906	2	0

Table 3.1: Training data.

Note that **Income** denotes the income of a customer in thousands, **Limit** denotes the credit limit while **Cards** stands for the number of credit cards that the customer has. Demonstrate your supervised learning skill by constructing a proper table for kNN with the Euclidean distance and $k = 2$ and $k = 4$ to find the balance of a customer with an Income of 11.187 thousands, a credit limit of 5099 and having 4 credit cards. (3 marks)

Ans. After calculating the distances of the point we want to predict to each point in the training data, we obtain the following table.

Income	Limit	Cards	dist	rank	Balance
71.408	7114	2	2015.9	5	872
98.515	8760	5	3662.04	8	1230
16.479	5435	2	336.05	2	937
104.483	7140	2	2043.13	6	583
27.578	2531	1	2568.05	7	0
53.217	4943	2	161.58	1	382
15.866	3085	1	2014.01	4	136
11.808	1300	3	3799	9	0
35.61	6135	4	1036.29	3	992
15.602	906	2	4193	10	0

..... [2.4 marks]

The prediction with kNN ($k = 2$) is 659.5 [0.3 mark]

The prediction with kNN ($k = 4$) is 611.75 [0.3 mark]

7. Given the Irish education transitions data with the following features

- Sex: gender of the Irish schoolchildren aged 11 in 1967 (female, male);
- DVRT: Drumcondra Verbal Reasoning Test Score (integers);
- PS: Prestige score (integers);
- TS: Type of school (primary_terminal_leaver, secondary, vocational);

and the labelled output is the Leaving.Certificate with two values — not.taken ($Y = 0$) or taken ($Y = 1$). Suppose the trained supervised learning logistic regression model has the following analysis result.

```
Call:
glm(formula = Leaving.Certificate ~ Sex + DVRT + PS + TS, family = binomial,
    data = data.train)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -21.87135   747.84837  -0.029   0.97667
Sexmale         0.17392    0.28777   0.604   0.54559
DVRT           0.04211    0.01051   4.007 6.16e-05 ***
PS             0.02375    0.00908   2.615  0.00891 **
TSsecondary    17.03423   747.84785   0.023   0.98183
TSvocational   13.78790   747.84798   0.018   0.98529
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 520.71  on 379  degrees of freedom
Residual deviance: 331.36  on 374  degrees of freedom
(20 observations deleted due to missingness)
AIC: 343.36

Number of Fisher Scoring iterations: 16
```

- Determine the odds ratio for the Leaving.Certificate of the prestige score when it is increment by 1 from some unknown value x and then determine the ordering of the probabilities of $Y = 1$ for a prestige score x and for a prestige score $x + 1$ (1 mark)
- For a male student with a Drumcondra Verbal Reasoning Test Score of 82 and a pretige score of 18 with a type school from vocational, calculate the conditional probability and the odds of the Leaving Certificate is taken given the conditions. (1 mark)
- For a male student with a Drumcondra Verbal Reasoning Test Score of 94 and a pretige score of 43 with a type school from vocational, calculate the conditional probability and predict whether the Leaving Certificate is taken or not given the conditions. (1 mark)

Ans.

(a)

$$\frac{Odds(PS = x + 1)}{Odds(PS = x)} = \exp(0.02374737) = 1.024032 \quad [0.5 \text{ mark}]$$

$$P(Y = 1|PS = x + 1) > P(Y = 1|PS = x) \quad [0.5 \text{ mark}]$$

item	Sexmale	DVRT	PS	TSsecondary	TSvocational	logit	prob	odds / $\hat{Y}$	mark
-21.8714	0.1739	0.0421	0.0237	17.0342	13.7879	NA	NA	NA	-
(b)	1	82	18	0	1	-4.029422	0.017474	0.017785	[1 mark]
(c)	1	94	43	0	1	-2.930472	0.050668	not.taken	[1 mark]

